

EXPERIMENT 6

AIM: Interfacing of component in project

THEORY:

Interfacing electronic components is a fundamental step in building any embedded system. In the context of this gas leakage detection system, component interfacing refers to the correct **electrical and logical connections** between various hardware parts—such as the **MQ5 gas sensor**, **Arduino Uno**, **LED**, and **supporting passive components**—so they work together as a functional unit.

The **main objective** of interfacing in this project is to enable smooth data flow from the sensor (input) to the microcontroller (Arduino), and then control the output devices (LEDs, buzzers, etc.) based on the logic programmed.

In this project:

- The **MQ5 sensor** is connected to the **analog input pin A0** of the Arduino to send continuous gas concentration data in the form of voltage.
- The **LED** is connected to a **digital output pin (e.g., pin 13)** to receive ON/OFF signals depending on whether the gas level has crossed the defined threshold.
- A **220Ω resistor** is placed in series with the LED to limit current and prevent damage.
- A **breadboard** and **jumper wires** are used to make all temporary connections safe and organized for prototyping and testing.

Understanding how each component is connected and how it interacts with others is essential for ensuring the reliability and functionality of the overall system.

Component Interfacing Summary:

Component	Interfacing Details & Pins	Notes
MQ5 Sensor	VCC → 5V (Arduino), GND → GND, AOUT → A0 (Analog input pin)	Sends analog voltage proportional to gas concentration
Arduino Uno	USB → PC for code & power, A0 for MQ5 input, D13 for LED output	Controls logic, processes data, and triggers output signals
LED	Anode → D13 (via 220Ω resistor), Cathode → GND	Turns ON/OFF based on gas level
Resistor	220Ω between D13 and LED Anode	Limits current to protect the LED
Breadboard	Central platform for making connections	Allows for easy testing and adjustments
Jumper Wires	Used for all pin connections	Ensures secure and flexible wiring between components and Arduino

CONCLUSION:

The interfacing of components in this project was successfully achieved, ensuring seamless communication between the sensor, microcontroller, and output indicators.

Key takeaways:

- Proper power supply and grounding are crucial for component functionality.
- Analog signals from the MQ5 sensor can be effectively read and processed by the Arduino.
- Digital output signals from the Arduino can successfully control external devices like LEDs.
- Using passive components (like resistors) appropriately helps protect sensitive components.